Requirement Analysis

Technology Stack (Architecture & Stack)

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, "Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau," focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Table 1 — Components & Technologies

Component	Technology Used	Purpose
Data Source	CSV (cosmetics product dataset)	Raw cosmetics product data (1,472 records, 11 attributes)
Data Cleaning	Python 3 (Pandas library)	Brand-name casing, ingredient-placeholder cleanup, zero-rank handling
Spreadsheet Handling	Microsoft Excel (openpyxl)	Storing and formatting the cleaned dataset (cosmetics_cleaned.xlsx)
Data Visualization	Tableau Desktop / Tableau Public	Building all chart types and the combined dashboard
Dashboard Publishing	Tableau Public (free cloud hosting)	Browser-accessible dashboard sharing via public URL
Documentation	Microsoft Word (.docx)	Project deliverables and phase documentation
Project Management	SkillWallet platform	Epic/story tracking across the team

Table 2 — Application Characteristics

Characteristic	Description
Open Source Tools	Python (Pandas) is open-source; Tableau Public is free for public publishing
Security	No PII data; dataset contains only publicly available cosmetics product information
Scalability	New products or brands can be appended to the CSV and the workbook refreshed without redesign
Availability	Tableau Public guarantees high availability; no server maintenance required
Performance	1,472 rows (7,360 post-pivot) render and filter within 2 seconds on a standard laptop
Compatibility	Works on Chrome, Edge, Safari; responsive on desktop and Tableau Public mobile view